Quiz 1 Solution Semester 1, 2024

The University of Sydney

Faculties of Arts, Economics, Education, Engineering and Science

MATH2921: Vector Calculus and Differential Equations (Advanced)

Lecturer: Zhou Zhang

Time allowed: 40-Minute In-Person Quiz

This booklet contains **7** pages.

Name (Last, First): ..... SID: .....

Total Points *(out of 20)*

- Open book. No internet. Individual Work. No discussion.
- Covering material in Weeks 1–5 lectures and practice classes (corresponding to Tutorial Sheets 1–5).
- Time and Venue: Week 7 Practice Class (Thursday, 11 April, 12NOON–1PM), Physics Lecture Theatre (1) 405.
- Style: 5 short answer questions with limited requirements in showing works. Each question with 4 points: 2 boxed answers for 1 point each, and another 1 point for relevant works for each.

1. Consider the surface $\{(x, y, z) \mid x^2 + y^2 + z^2 = 1\}$ as a graph of $x = f(y, z)$ near the point $(-1, 0, 0)$, then

i) $f(\cancel{x, y}\ y, z) = \boxed{-\sqrt{1-y^2-z^2}}$;

ii) at the point $(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0)$, the normal vector of this surface with the x-component being 1 is $(1, a, b)$. So $a + b = \boxed{-1}$.

i) Near $(-1,0,0)$, $x<0$.

$$x = -\sqrt{1-y^2-z^2}$$

ii) Level surface $\underbrace{x^2+y^2+z^2-1}_{F(x,y,z)}=0$ $\quad \nabla F = (2x, 2y, 2z)$

At $(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0)$, $\nabla F = (\sqrt{2}, -\sqrt{2}, 0) \xrightarrow{1/\sqrt{2}} (1, \underset{a}{-1}, \underset{b}{0})$

2. Consider the function $f(x, y) = x^2 + y^2 + 2x + 2y$ over the $2D$ region $R = \{(x, y) \mid x \in [-2, 0], y \in [-2, 0]\}$.

i) The only critical point is $\boxed{(-1, -1)}$. ← no need to classify

ii) The global maximum and minimum of $f(x, y)$ in R add up to $\boxed{-2}$.

i) $\begin{cases} f_x = 2x+2 = 0 \\ f_y = 2y+2 = 0 \end{cases}$ $\quad x=-1,\ y=-1.$ $\quad (-1,-1)$

ii) Compare $(-1,-1)$ and ∂R. $f(x,y) = (x+1)^2+(y+1)^2-2$

$f(-1,-1) = \underline{-2}$ (clearly global $\underline{\text{minimum}}$)

The boundary is x or $y = 0$ or -2 with the other one in $[-2, 0]$. Just consider

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$g(t) = 1^2 + (t+1)^2 - 2 = (t+1)^2 - 1$ for $t \in [-2, 0]$.

Max is only possible for $t = -2, 0$. $g(0) = g(-2) = 0$, global $\underline{\text{max}}$ $\underline{0}$.

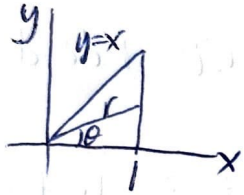
3. For the $2D$ region $R = \{(x, y) \mid y \le x, x \le 1, y \ge 0\}$,

i) express R in polar coordinates: [ ]

ii) rewrite the double integral $\iint_R f(x, y)dxdy$ as an iterated integral with dx inside and dy outside (i.e. using the y-first description of R):

[ ]

i)



$\theta \in [0, \pi/4]$.

$\cos\theta = 1/r$.

$r = 1/\cos\theta$

polar coordinates

$R = \{(r, \theta) \mid 0 \le \theta \le \pi/4,\ 0 \le r \le 1/\cos\theta\}$

ii)

$R = \{(x, y) \mid 0 \le y \le 1,\ y \le x \le 1\}$

$\iint_R f(x,y)\,dx\,dy$

$= \int_0^1 \int_y^1 f(x,y)\,dx\,dy$

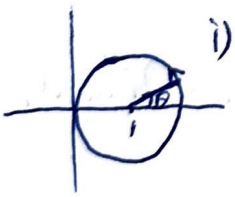
4. Consider the vector field $\vec{F}(x, y) = (y, 0)$ and the curve C ~~which~~ is ~~the unit circle~~ centred at $(1, 0)$ in the counter-clockwise direction.

i) The total work done by the force field $\vec{F}$ along C is equal to [$-\pi$] (the numerical answer, NOT the line integral).

ii) The double integral from Green's Theorem for the line integral in i) is

[ ] $-\iint_R dxdy$, $R = \{(x,y) \mid (x-1)^2 + y^2 \le 1\}$

(just the double integral expression with the explicit integrand function and explicit $2D$ region).



i) $C: \vec{r}(\theta) = (\overset{x(\theta)}{1+\cos\theta}, \overset{y(\theta)}{\sin\theta})$, $\theta \in [0, 2\pi)$

$\int_C \vec{F}\cdot d\vec{r} = \int_0^{2\pi} (\sin\theta, 0)\cdot(-\sin\theta, \cos\theta)\,d\theta$

$= -\int_0^{2\pi} \sin^2\theta\,d\theta = -\frac{1}{2}\int_0^{2\pi}(1 - \cos 2\theta)\,d\theta$

$= -\frac{1}{2}\left(\theta - \frac{1}{2}\sin 2\theta\right)\Big|_0^{2\pi}$

$= -\pi$.

ii) $R = \{(x,y) \mid (x-1)^2 + y^2 \le 1\}$ unit disk

$\oint_C \vec{F}\cdot d\vec{r} = \iint_R \left(\frac{\partial 0}{\partial x} - \frac{\partial y}{\partial y}\right)dxdy = -\iint_R dxdy$

$= -\pi$

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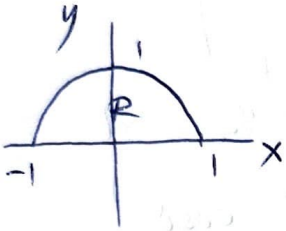
5. A metal plate is in the shape of the $2D$ region $R = \{(x, y) \mid x^2 + y^2 \leq 1, y \geq 0\}$. The (varying) density is $\rho(x, y) = y + x^2$.

i) Its total mass is $\boxed{\frac{2}{3} + \frac{\pi}{8}}$.

ii) Its centre of mass is at $\boxed{}$ $\leftarrow \left(0, \dfrac{\frac{\pi}{8} + \frac{2}{15}}{\frac{\pi}{8} + \frac{2}{3}}\right)$



$R = \left\{(r, \theta) \,\middle|\, \begin{array}{l} r \leq 1 \\ 0 \leq \theta \leq \pi \end{array}\right\}$

i) $\iint_R (y + x^2)\,dx\,dy = \int_0^1 \int_0^\pi (r\sin\theta + r^2\cos^2\theta)\, r\, d\theta\, dr$

$= \int_0^1 \int_0^\pi (r^2\sin\theta + r^3\cos^2\theta)\, d\theta\, dr$

$= \int_0^1 r^2 dr \int_0^\pi \sin\theta\, d\theta + \int_0^1 r^3 dr \int_0^\pi \cos^2\theta\, d\theta$

$= \frac{1}{3} \times \left(-\cos\theta \,\middle|_0^\pi\right) + \frac{1}{4} \times \int_0^\pi \underbrace{\frac{1}{2}(1 + \cos 2\theta)}_{\text{"}\sin^2\theta\text{"}}\, d\theta$

$= \frac{1}{3} \times (-(-1) - (-1)) + \frac{1}{4} \times \frac{\pi}{2}$

$= \frac{2}{3} + \frac{\pi}{8}.$

ii) $\iint_R x(y + x^2)\,dx\,dy$

$= \int_0^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \underbrace{yx + x^3}_{\text{odd for } x}\, dx\, dy$

$= 0$

$\iint_R y(y + x^2)\,dx\,dy = \int_0^1 \int_0^\pi (r^2\sin^2\theta + r^3\sin\theta\cos^2\theta)\, r\, d\theta\, dr$

$= \int_0^1 r^3 dr \cdot \int_0^\pi \sin^2\theta\, d\theta + \int_0^1 r^4 dr \cdot \int_0^\pi \underbrace{\sin\theta\cos^2\theta}_{\text{"}\sin\theta\text{"}}\, d\theta$

$= \frac{1}{4} \times \int_0^\pi \frac{1}{2}(1 - \cos 2\theta)\, d\theta$

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$+ \frac{1}{5} \times (-1) \int_0^\pi \cos^2\theta\, d\cos\theta$

$u = \cos\theta$

$= \frac{1}{4} \times \frac{\pi}{2} - \frac{1}{5} \int_1^{-1} u^2 du = \frac{\pi}{8} + \frac{2}{15}$